Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

Three solenoid coils of same dimension, same number of turns and same number of layers of windings are taken. Coil 1 with inductance $\mathrm{L_1}$ was wound using a Mn wire of resistance $11\Omega/\mathrm{m}$, coil 2 with inductance $\mathrm{L_2}$ was wound using the similar wire but the direction of winding was reversed in each layer; coil 3 with inductance $\mathrm{L_3}$ was wound using a superconducting wire. The self inductance of the coils $\mathrm{L_1,\ L_2,\ L_3}$ are

(A) $\mathrm{L_1 = L_2 = L_3}$

(B) $\mathrm{L_1 = L_2; L_3 = 0}$

(C) $\mathrm{L_1 = L_3; L_2 = 0}$

(D) $L_1 > L_2 > L_3$

Q.2

The mutual inductance of a pair of coils, each of N turns, is M henry. If a current of I ampere in one of the coils is brought to zero in t second, the emf induced per turn in the other coil, in volt, will be

(A) $\frac{\mathrm{MI}}{\mathrm{t}}$

(B) $\frac{\mathrm{NMI}}{\mathrm{t}}$

(C) $\frac{\mathrm{MN}}{\mathrm{It}}$

(D) $\frac{\mathrm{MI}}{\mathrm{Nt}}$

Q.3

In an inductor of self-inductance $L = 2\mathrm{mH}$, current changes with time according to relation $\mathrm{i} = \mathrm{t}^2\mathrm{e}^{-\mathrm{t}}$. At what time (in second) emf is zero?

(A) 1

(B) 2

(C) 3

(D) 4

Q.4

Two coaxial solenoids are made by winding thin insulated wire over a pipe of cross-sectional area $\mathrm{A} = 10\ \mathrm{cm}^2$ and length $= 20\ \mathrm{cm}$. If one of the solenoid has 300 turns and the other 400 turns, then find their mutual inductance (in Henry) is:
$\left(\mu_0 = 4\pi \times 10^{-7}\mathrm{TmA}^{-1}\right)$
(Take pi as 3.14)

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(A) 0.007536

(B) 0.07536

(C) 0.0007536

(D) 0.00007536

Q.5

The approximate formula expressing the formula of mutual inductance of two thin coaxial loops of the same radius a when their centers are separated by a distance l with $l >> \mathrm{a}$ is

(A) $\frac{1}{2}\frac{\mu_0 \pi \mathrm{a}^4}{l^3}$

(B) $\frac{1}{2}\frac{\mu_0 \pi \mathrm{a}^4}{l^2}$

(C) $\frac{\mu_0}{4\pi}\frac{\pi a^4}{l^2}$

(D) $\frac{\mu_0}{\pi}\frac{\mathrm{a}^4}{l^3}$

Q.6

Two coils are at fixed locations. The coil 1 has no current and the current in coil 2 increases at the rate of $15.0\ \mathrm{A\ s^{-1}}$, the emf in coil 1 is $25\mathrm{mV}$, when coil 2 has no current and coil 1 has a current of $3.6\ \mathrm{A}$, the flux linkage on coil 2 is

(A) $16\ \mathrm{m\ Wb}$

(B) $10\ \mathrm{m\ Wb}$

(C) $4.00\ \mathrm{m\ Wb}$

(D) $6.00\mathrm{mWb}$

Q.7

What is the self-inductance of a coil which produces $5\ \mathrm{V}$ when the current changes from 3 ampere to 2 ampere in one millisecond?

(A) 5000 henry

(B) 5 milli-henry

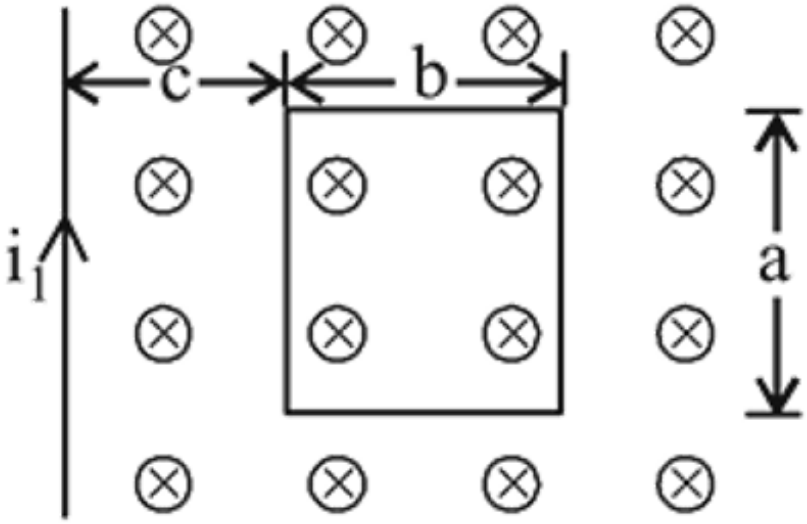
(C) 50 henry

(D) 5 henry

Q.8

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Figure shows a rectangular coil near a long wire. The mutual inductance of the combination is



(A) $\frac{\mu_0 a}{2\pi} \ln\left(1 - \frac{b}{c}\right)$

(B) $\frac{\mu_0 a}{2\pi} \ln\left(1 + \frac{b}{c}\right)$

(C) $\frac{\mu_0 a}{\pi} \ln\left(1 + \frac{b}{c}\right)$

(D) $\frac{\mu_0 a}{\sqrt{2}\pi} \ln\left(1 + \frac{b}{c}\right)$

Q.9

In an inductor of self-inductance $L = 2\text{mH}$, current changes with time according to relation $i = t^2 e^{-t}$. At what time emf is zero?

(A) 4 s

(B) 3 s

(C) 2 s

(D) 1 s

Q.10

A varying current in a coil changes from 10 A to zero in 0.5sec. If the average e.m.f induced in the coil is 220 V, the self-inductance of the coil is

(A) 5H

(B) 6H

(C) 11H

(D) 12H

Q.11

When the number of turns in a coil is doubled without any change in the length of the coil, its self-inductance becomes

(A) four times

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(B) doubled

(C) halved

(D) squared

Q.12

A capacitor of capacitance $2\mu\mathrm{F}$ is charged to a potential difference of $12V$. The charging battery is then removed and the capacitor is connected across an inductor of self inductance $0.6\mathrm{mH}$. The current in the circuit at a time when the potential difference across the capacitor is $6 \vee$ is

(A) $0.3A$

(B) $0.6\ \mathrm{A}$

(C) $0.9\ \mathrm{A}$

(D) $1.2\ \mathrm{A}$

Q.13

A fully charged capacitor C with initial charge q_0 is connected to a coil of self inductance L at $t = 0$. The time at which the energy is stored equally between the electric and the magnetic fields is:

(A) $\frac{\pi}{4}\sqrt{LC}$

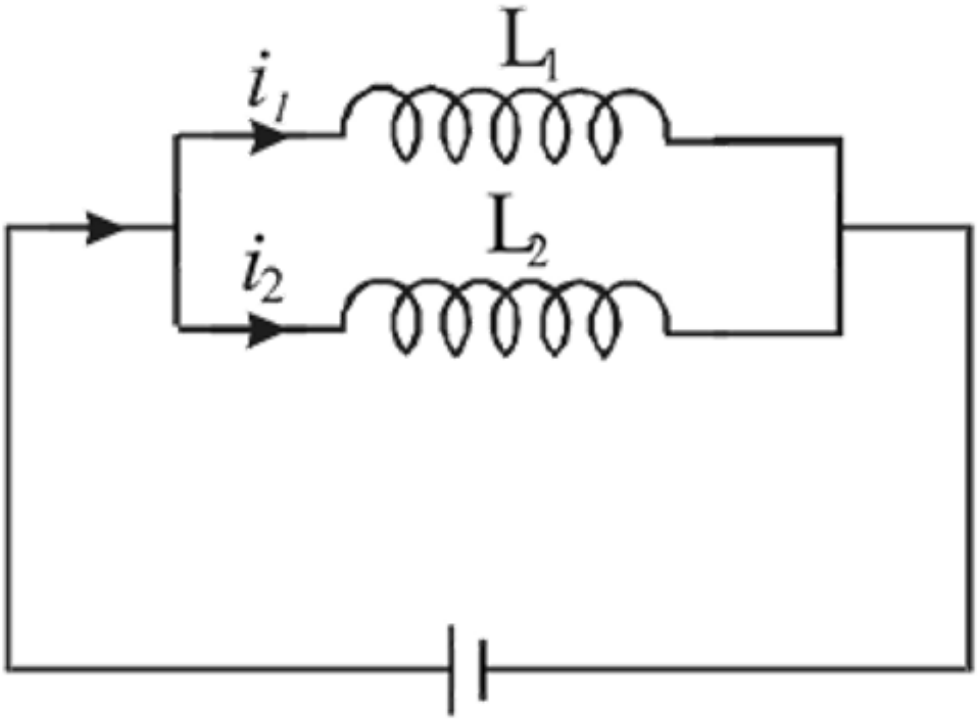
(B) $2\pi\sqrt{LC}$

(C) $\sqrt{LC}$

(D) $\pi\sqrt{LC}$

Q.14

Two inductances L_1 and L_2 are placed closer and in parallel. Their combined inductance is



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(A) $\frac{L_1 L_2}{L_1+L_2}$

(B) $(L_1 + L_2)$

(C) $(L_1 + L_2)\frac{L_1}{L_2}$

(D) $(L_1 + L_2)\frac{L_2}{L_1}$

Q.15

The magnetic flux linked with a coil is ϕ and the EMF induced in it is E, then

(A) If $\phi = 0$, E must be 0

(B) If $\phi \neq 0$, E cannot be zero

(C) If E is not 0, ϕ may or may not be 0

(D) none of the above is correct

Answers & Solutions

Q.1 Answer:

$L_1 = L_2; L_3 = 0$

Solution:

The self-inductance of a coil $L = \mu_0 n^2 Al$ where μ_0 = permeability of air, n = number of turns per unit length, A = Area of cross-section and l = length of the solenoid This depends on the geometry of the inductor such as cross-sectional area, length and number of turns and not on the material, even if it is made of a super conducting material. If the superconductor is below the critical temperature, the current will continuously flow and the inductance may not have the property of inductance any more.

$\therefore L_1 = L_2; L_3 = 0.$

Q.2 Answer:

$\frac{MI}{t}$

Solution:

$E = \frac{d}{dt}(NMI) \Rightarrow E = NM\frac{dI}{dt} \Rightarrow E = \frac{NMI}{t}$

emf induced per unit turn $= \frac{E}{N} = \frac{MI}{t}$

Q.3 Answer:

2

Solution:

$L = 2\text{mH}, i = t^2 e^{-t}$

$E = -L\frac{di}{dt} = -L\left[-t^2 e^{-t} + 2te^{-t}\right]$

when $E = 0$

$-e^{-t}t^2 + 2te^{-t} = 0$

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or, $2te^{-t} = e^{-t}t^2$

$\Rightarrow t = 2\text{sec}$

Q.4 Answer:

0.0007536

Solution:

$\mathrm{M} = \frac{\mu_0 \, \mathrm{N}_1 \, \mathrm{N}_2 \, \mathrm{A}}{\ell} = \frac{4\pi \times 10^{-7} \times 300 \times 400 \times 100 \times 10^{-4}}{0.2}$

$= 2.4\pi \times 10^{-4}\mathrm{H}$

Q.5 Answer:

$\frac{1}{2}\frac{\mu_0 \pi \mathrm{a}^4}{l^3}$

Solution:

Let I = current in one loop. The magnetic field at the centre of the other co-axial loop at a distance 1 from the centre of the first loop is

$\mathrm{B} = \frac{\mu_0}{4\pi}\frac{2\mathrm{I}\pi\mathrm{a}^2}{(\mathrm{a}^2+1^2)^{3/2}}$

where, $\mathrm{P_m} = \mathrm{I}\pi\mathrm{a}^2$ **= magnetic moment of the loop**

Flux through the other loop is

$\phi_{12} = \mathrm{B}\pi\mathrm{a}^2 = \frac{\mu_0}{4\pi}\frac{2\pi\mathrm{a}^2\mathrm{I}}{(\mathrm{a}^2+1^2)^{3/2}}\pi\mathrm{a}^2$

$\text{or } \mathrm{M} = \frac{\phi_{12}}{\mathrm{I}} = \frac{\mu_0}{4\pi}\frac{2\pi^2\mathrm{a}^4}{(\mathrm{a}^2+1^2)^{3/2}}$

$= \frac{\mu_0 \pi a^4}{2(a^2+1^2)^{3/2}} = \frac{\mu_0 \pi a^4}{21^3}(a << 1)$

Q.6 Answer:

$6.00\mathrm{mWb}$

Solution:

$|\mathrm{M}| = \frac{\mathrm{e}_1}{(\mathrm{di}_2/\mathrm{dt})} = \frac{\phi_2}{\mathrm{i}_1}$

$\therefore \phi_2 = \frac{\mathrm{e}_1 \mathrm{i}_1}{(\mathrm{di}_2/\mathrm{dt})} = \frac{\left(25.0 \times 10^{-3}\right)(3.6)}{(15)}$

$= 6 \times 10^{-3} = 6\mathrm{mWb}$

Q.7 Answer:

5 milli-henry

Solution:

$L = \frac{e}{\frac{di}{dt}} = \frac{edt}{di} = \frac{5 \times 10^{-3}}{(3-2)}\mathrm{H} = 5\mathrm{mH}$

Q.8 Answer:

$\frac{\mu_0 \mathrm{a}}{2\pi}\ln\left(1 + \frac{\mathrm{b}}{\mathrm{c}}\right)$

Solution:

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Let current $\mathrm{i_1}$ in straight wire be upward. Then the magnetic field due to the straight wire has magnitude $\mathrm{B_1 = \mu_0 i_1/2\pi r}$ at distance r. In accrodance with right hand rule, $\mathrm{B_1}$ points inward to the plane of page. We consider a differential strip of thickness dr, area $\mathrm{dA_2 = adr}$. Magnetic flux through area dA, $\mathrm{d\phi_B = B_1^2(adr)}$.

Total flux through the loop,

$$\begin{aligned}\mathrm{\phi_B} &= \int \mathrm{B_1 adr} = \int_{\mathrm{c}}^{\mathrm{c+b}} \frac{\mathrm{\mu_0 i_1}}{\mathrm{2\pi r}} \mathrm{adr} \\ &= \frac{\mathrm{\mu_0 i_1 a}}{\mathrm{2\pi}} \int_{\mathrm{c}}^{\mathrm{c+b}} \frac{\mathrm{dr}}{\mathrm{r}} = \frac{\mathrm{\mu_0 i_1 a}}{\mathrm{2\pi}} \ln\left(\frac{\mathrm{c+b}}{\mathrm{c}}\right)\end{aligned}$$

Therefore mutual inductance,

$$\mathrm{M} = \frac{\phi}{\mathrm{i_1}} = \frac{\mathrm{\mu_0 a}}{\mathrm{2\pi}} \ln\left(1 + \frac{\mathrm{b}}{\mathrm{c}}\right)$$

Q.9 Answer:

$2\ \mathrm{s}$

Solution:

$L = 2\mathrm{mH}, i = t^2\mathrm{e}^{-t}$

$E = -L\frac{di}{dt} = -L\left[-t^2e^{-t} + 2te^{-t}\right]$

when $E = 0, -e^{-t}t^2 + 2te^{-t} = 0$

or, $2te^{-t} = e^{-t}t^2 \Rightarrow t = 2\mathrm{sec}.$

Q.10 Answer:

$11\mathrm{H}$

Solution:

Initial current $(\mathrm{I_1}) = 10\ \mathrm{A}$; **Final current** $(\mathrm{I_2}) = 0$; **Time** $(\mathrm{t}) = 0.5\mathrm{sec}$ **and induced e.m.f.** $(\varepsilon) = 220$

V. Induced e.m.f. (\varepsilon) $= -L\frac{dI}{dt} = -L\frac{(I_2 - I_1)}{t} = -L\frac{(0-10)}{0.5} = 20\ \mathrm{L}$

or, $L = \frac{220}{20} = 11\mathrm{H}$

[where L **= Self inductance of coil]**

Q.11 Answer:

doubled

Solution:

$L = \frac{\mu_0 N^2 \pi r}{2}$

When the number of turns in a coil is doubled without changing the length of the coil, the radius of the coil is half.

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Hence, $L' = \frac{\mu_0 (2N)^2 \times \pi (r/2)}{2} = 2\left[\frac{\mu_0 N^2 \pi r}{2}\right] = 2L$

$\therefore$ **(b)**

Q.12 Answer:

0.6 A

Solution:

We know that $\mathrm{Q} = \mathrm{CV}$ **and** $\mathrm{Q} = \mathrm{Q}_0 \cos \omega \mathrm{t}$. **Also** $\mathrm{Q}_0 = \mathrm{CV}_0$

$\therefore \cos \omega \mathrm{t} = \frac{\mathrm{Q}}{\mathrm{Q}_0} = \frac{\mathrm{V}}{\mathrm{V}_0} = \frac{6}{12} = \frac{1}{2}$ **or** $\omega \mathrm{t} = \frac{\pi}{3}$.

Now ω **is given by** $\omega = \frac{1}{\sqrt{LC}}$ **(1)**

Given $L = 0.6 \times 10^{-3} \mathrm{H}$ **and** $\mathrm{C} = 2 \times 10^{-6}$ F. **Using these values in Eq. (1)**

we get $\omega = \frac{10^5}{2\sqrt{3}} \mathrm{rads}^{-1}$

Now $\mathrm{I} = \frac{\mathrm{dQ}}{\mathrm{dt}} = \frac{\mathrm{d}}{\mathrm{d}} \left(\mathrm{Q}_0 \cos \omega \mathrm{t}\right) = -\mathrm{Q}_\mathrm{n} \omega \sin \omega \mathrm{t}$

$\therefore |\mathrm{I}| = \mathrm{Q}_0 \omega \sin \omega \mathrm{t} = \mathrm{CV}_0 \omega \sin \omega \mathrm{t} = \left(2 \times 10^{-6}\right) \times 12 \times \frac{10^5}{2\sqrt{3}} \sin\left(\frac{\pi}{3}\right) = 0.6$ A

Q.13 Answer:

$\frac{\pi}{4}\sqrt{LC}$

Solution:

Energy stored in magnetic field $= \frac{1}{2}\mathrm{Li}^2$

Energystored in electric field $= \frac{1}{2}\frac{q^2}{C}$

$\therefore \frac{1}{2}Li^2 = \frac{1}{2}\frac{q^2}{C}$

Also $q = q_0 \cos \omega t$ **and** $\omega = \frac{1}{\sqrt{LC}}$

On solving $t = \frac{\pi}{4}\sqrt{LC}$

Q.14 Answer:

$\frac{\mathrm{L}_1 \mathrm{L}_2}{\mathrm{L}_1 + \mathrm{L}_2}$

Solution:

Parallel $= \frac{L_1 L_2}{L_1 \times L_2}$

Q.15 Answer:

If E is not 0, ϕ may or may not be 0

Solution:

Induced emf,

$E = -\frac{d\phi}{dt}$

Therefore, $E = 0$ **when** $\frac{d\phi}{dt} = 0$ **and vice versa.**

Even if $\phi = 0, \frac{d\phi}{dt}$ **may be non-zero.**

Similarly, if $\phi \neq 0, \frac{d\phi}{dt}$ **may be zero.**

For example, if $\phi = \phi_0 \sin \omega t$

Then, $E = -\frac{d\phi}{dt} = -\phi_0 \cos \omega t$

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